

PAIRED CLIQUE MINORS FROM CONNECTED REGIONS

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ABSTRACT. Let R, S be disjoint sets of k vertices joined by k vertex-disjoint paths. We prove that $k - 1$ disjoint connected regions, each adjacent to every vertex of $R \cup S$ and avoiding the terminals, force a K_k -minor whose branch sets each contain one vertex of R and one of S . The pairing is not prescribed, and the number of regions is best possible. In a stronger one-sided theorem, the regions cover S and each meets S , while adjacency is required only to R . A chain of minimum vertex separators shows that a linkage in a reduced graph has pairwise adjacent paths. The reductions give a polynomial-time construction when the regions are supplied.

1. STATEMENTS

All graphs are finite and simple. For disjoint vertex sets R, S , an R - S *linkage of order k* is a family of k pairwise vertex-disjoint paths with distinct ends in R and in S . We write $\kappa_G(R, S)$ for the largest possible order. A vertex set is *full to R* if every vertex of R has a neighbour in it. A *region* is a nonempty vertex set inducing a connected subgraph.

For disjoint k -element sets R, S , a *paired clique model* consists of k pairwise disjoint nonempty connected branch sets that are pairwise adjacent, each containing exactly one vertex of R and exactly one vertex of S . This is a K_k -minor model with a free terminal pairing. The regions in the following statements need not be preserved as additional branch sets.

Theorem 1.1. *Let $k \geq 2$, let R, S be disjoint k -element vertex sets in a graph G , and suppose that $\kappa_G(R, S) \geq k$. Suppose there are $k - 1$ pairwise disjoint regions $H_1, \dots, H_{k-1}$ avoiding R , each full to R , such that*

$$S \subseteq \bigcup_{j=1}^{k-1} H_j \quad \text{and} \quad S \cap H_j \neq \emptyset \quad (1 \leq j < k).$$

Then G has a paired clique model for R, S .

The theorem requires no global connectivity assumption on G and implies the following symmetric statement.

Corollary 1.2. *Let $k \geq 1$ and let R, S be disjoint k -element vertex sets in G . Suppose G contains $k - 1$ pairwise disjoint regions avoiding $R \cup S$,*

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each full to $R \cup S$. Then G has a paired clique model for R, S if and only if $\kappa_G(R, S) \geq k$.

We prove Theorem 1.1 in Sections 2 through 5. Section 6 gives the polynomial-time construction and shows that the number of regions in Corollary 1.2 cannot be reduced below $k - 1$, for any $k \geq 2$.

In the terminology of Protopapas, Thilikos and Wiederrecht [2, Sections 1.1 and 2.1], the conclusion is a *rainbow K_k -minor* when R and S are the two colour sets and all other vertices are uncoloured. Each branch set meets both colour sets exactly once, since each set has k vertices. Their Theorem 3.1, specialised to two colours and a rainbow K_k , starts from a K_{4k} -minor and gives a rainbow minor or a separator outcome. Our sufficient condition uses the supplied connected regions and the linkage. We make no claim of publication priority.

An R - S *separator* is a set $C \subseteq V(G)$ such that $G - C$ has no path from $R \setminus C$ to $S \setminus C$; terminal vertices are allowed in C . We use the vertex form of Menger's theorem [1]: the minimum size of such a separator equals $\kappa_G(R, S)$. In particular, when $|R| = |S| = k$ and a linkage has order k , every terminal is a path end and no terminal is internal to a linkage path.

2. REDUCTIONS AND TERMINAL OWNERSHIP

Suppose Theorem 1.1 is false. Choose a counterexample with k minimum and, subject to this, with $|V(G)|$ minimum. Since k vertices of S occupy $k - 1$ regions, each meeting S , exactly one region contains two of them and every other region contains one.

No region is a singleton. Indeed, a singleton region is $\{s\}$ for some $s \in S$. Fix a linkage of order k and let $r \in R$ be the other end of its path ending at s . Delete r, s . The other $k - 1$ paths form a linkage for $R \setminus \{r\}, S \setminus \{s\}$, and the remaining $k - 2$ regions satisfy the hypotheses of the theorem for $k - 1$. Here $k \geq 3$: when $k = 2$, the only region contains both vertices of S . Minimality of k supplies a paired clique model in $G - \{r, s\}$. Add the branch set $\{r, s\}$. It is connected, since $\{s\}$ is full to R , and s is adjacent to the R vertex in each other branch set. The earlier branch sets avoid r, s , since both terminals were deleted before induction.

We may assume that the regions cover $V(G) \setminus R$. To see this, consider a component D of $G - (R \cup \bigcup_j H_j)$. If D contacts a region, absorb it into that region. This preserves disjointness, connectivity, fullness and all terminal memberships. If it contacts no region, its neighbourhood is contained in R . Every linkage of order k avoids D : a path visiting D would have to pass through an R terminal internally to reach its S end. Deleting D therefore preserves the hypotheses, contrary to minimum host order. Absorbing all remaining components gives

$$(1) \quad V(G) = R \dot{\cup} H_1 \dot{\cup} \cdots \dot{\cup} H_{k-1}.$$

The graph has not changed, and no region is a singleton.

Call an edge within a region *permitted* if its ends are not both in S . Contracting a permitted edge preserves the two terminal sets: an S endpoint, if present, retains its name. The resulting regions are still disjoint and connected, full to R , and have the same S memberships. If the quotient has a linkage of order k , minimum host order supplies its paired clique model. Lift that model by replacing the contracted vertex, if used, by its connected two-vertex preimage. This preimage contains no R vertex and at most one S vertex. The lift preserves disjointness, all contacts and every terminal assignment. Consequently,

$$(2) \quad \kappa_{G/uv}(R, S) \leq k - 1 \quad \text{for every permitted region edge } uv.$$

3. SEPARATORS AND A SPANNING LINKAGE

Lemma 3.1. *For every permitted edge uv in H_j , there is an R – S separator of size k containing u, v , exactly one vertex in each other region, and no vertex of R . For any fixed linkage of order k , these k separator vertices lie one on each path.*

Proof. By (2) and Menger’s theorem, G/uv has a separator C' of size at most $k - 1$. It contains the contracted vertex, since otherwise C' would also separate R from S in G . Replace that vertex by u, v to obtain a separator C in G of size at most k . The existing linkage forces $|C| = k$.

Since $u, v \notin R$, the cut C removes at most $k - 2$ vertices of R , so some R vertex survives. If C missed another region, that region would connect this vertex to one of its S terminals in $G - C$. The remaining $k - 2$ cut vertices must therefore lie one in each of the other $k - 2$ regions. In particular, C avoids R . Each of the k disjoint linkage paths meets C , so $|C| = k$ gives exactly one meeting per path. $\square$

Fix a linkage $P_1, \dots, P_k$ and orient each path from its R end to its S end. This linkage spans G : every region vertex outside S has a neighbour in its region because the region is connected and non-singleton. The incident edge is permitted, and Lemma 3.1 places that vertex on the linkage. All terminals are already path ends, so (1) proves the claim. The ends of every region edge lie on different paths: use Lemma 3.1 for a permitted edge, and the distinct S ends for an edge with both ends in S .

4. A CHAIN OF SEPARATORS

Write

$$P_i = p_{i,0}p_{i,1} \cdots p_{i,\ell_i}, \quad p_{i,0} \in R, \quad p_{i,\ell_i} \in S.$$

Consider all vectors $c = (c_1, \dots, c_k)$ with $1 \leq c_i \leq \ell_i$ such that $\{p_{i,c_i} : 1 \leq i \leq k\}$ is an R – S separator. Identify each vector with its cut and order the vectors coordinatewise. The cuts exclude R endpoints but may contain S endpoints. The vector $(\ell_1, \dots, \ell_k)$ selects S and is the maximum of the collection.

Since the linkage spans G , a vector c is separating exactly when there is no edge from a strict prefix $\{p_{i,t} : t < c_i\}$ to a strict suffix $\{p_{j,t} : t > c_j\}$. Every prefix vertex connects to its R end without meeting the selected cut, and every suffix vertex connects to its S end. A crossing edge would therefore give a path avoiding the cut; without one, no surviving path can pass from a prefix to a suffix. This remains true when a selected S endpoint has an empty suffix.

The separating vectors are closed under coordinatewise minimum and maximum, denoted by $\wedge$ and $\vee$. For a crossing edge at $c \wedge d$, its prefix end is before both original cuts. Choose the original cut attaining the minimum at the suffix end's coordinate; the edge crosses that original cut, a contradiction. For $c \vee d$, choose the original cut attaining the maximum at the prefix end's coordinate. Its suffix end is after both original cuts, giving the same contradiction.

Choose a maximal chain of separating vectors,

$$c^0 < c^1 < \dots < c^m.$$

It includes the minimum and maximum of this finite lattice. Every region vertex occurs in a cut on the chain. For a vertex outside S , Lemma 3.1 provides a cut containing it; for an S vertex, use the maximum cut. If a vertex $z = p_{i,t}$ were skipped, the extreme cuts imply that two consecutive chain cuts $c < d$ would satisfy $c_i < t < d_i$. A cut a containing z would then give the strictly intermediate separating vector

$$(a \vee c) \wedge d,$$

contrary to maximality of the chain. Monotonicity shows that the indices of chain cuts containing z form a nonempty interval I_z of integers.

Lemma 4.1. *For every edge uv within a region, $I_u \cap I_v \neq \emptyset$.*

Proof. If $u, v \in S$, their intervals contain the maximum cut. Otherwise Lemma 3.1 gives a cut a containing both ends, which lie on different paths. Suppose the occurrence intervals are disjoint. After interchanging the ends, let r be the last index in I_u and l the first in I_v , with $r < l$. At c^{r+1} , the vertex u is in a strict prefix. The actual edge uv prevents v from being in a strict suffix, so $l = r + 1$: this next cut contains v .

The separating vector

$$(a \vee c^r) \wedge c^{r+1}$$

lies between the two consecutive cuts and contains both u and v . It differs from c^r at v 's coordinate and from c^{r+1} at u 's coordinate, contradicting maximality. This step uses the edge uv to exclude v from a strict suffix when u enters a strict prefix. $\square$

5. THE FIRST CUT GIVES ALL CLIQUE CONTACTS

Every chain cut meets every region. Indeed, the cut avoids R ; an untouched region would contain an uncut S vertex and connect it to R by

fullness. A cut has k vertices and there are $k - 1$ regions, so each region contributes at most two vertices to a chain cut.

Fix a region H_j and form the intersection graph T_j of the intervals $(I_z : z \in H_j)$. No three of these intervals contain a common index. The graph T_j is a forest: if it had a cycle, choose on that cycle an interval with smallest right endpoint. Both its cycle neighbours contain that endpoint, because each meets the chosen interval and has a right endpoint no smaller. This would give three intervals with a common index.

By Lemma 4.1, the connected graph $G[H_j]$ is a spanning subgraph of T_j . A forest with a connected spanning subgraph is a tree and cannot have any further edge. Hence $G[H_j] = T_j$. Thus two vertices of one region that occur in a common chain cut are adjacent in G .

For each i , put $f_i = p_{i,1}$, the first vertex after the R end of P_i ; it may be an S end. Each f_i occurs in a separating vector. The coordinatewise minimum of one such vector for each i is therefore

$$F = \{f_1, \dots, f_k\}.$$

This is the minimum separating vector, so it is on the chain. Every region neighbour of an R terminal lies in F . A neighbour later on its path would have a suffix to S avoiding F , giving an R - S path avoiding this cut.

Exactly one region is represented twice in F , and every other region once. If f_i, f_j lie in the same region, the preceding tree argument gives the edge $f_i f_j$. If they lie in different regions, at least one of those regions is represented only once in F . The R endpoint of the other path is full to that region, and all its neighbours there lie in F , so it is adjacent to that unique representative. Thus P_i, P_j are adjacent in every case.

The vertex sets of $P_1, \dots, P_k$ form the required paired clique model, a contradiction. This proves Theorem 1.1. The singleton step decreases k after deleting both reserved terminals. Contraction decreases host order and lifts through a fixed connected preimage containing at most one terminal. Both steps preserve terminal ownership. $\square$

6. THE SYMMETRIC STATEMENT, SHARPNESS AND CONSTRUCTION

Proof of Corollary 1.2. A paired clique model supplies a linkage by choosing an R - S path inside each branch set. Conversely, suppose a linkage of order k exists. For $k = 1$, take its only path as the branch set. For $k \geq 2$, assign the k vertices of S to the $k - 1$ given regions, at least one to each, and adjoin each vertex to its assigned region. Fullness to S makes the enlarged regions connected. They remain pairwise disjoint, avoid R and are full to R . They cover S and each meets S , so Theorem 1.1 applies. $\square$

Proposition 6.1. *For every $k \geq 2$, Corollary 1.2 fails if its number of full regions is reduced to $k - 2$.*

Proof. Take k disjoint edges $r_i s_i$ with $R = \{r_1, \dots, r_k\}$ and $S = \{s_1, \dots, s_k\}$, and add $k - 2$ vertices, each adjacent to all $2k$ terminals. Add no other edges.

The matching is a linkage of order k , and each added vertex is a full singleton region avoiding the terminals.

In any proposed paired clique model, at least two of its k branch sets contain no added vertex. Each of these consists of exactly one R vertex and one S vertex. Connectivity forces it to be one of the matching edges, and two distinct such branch sets are not adjacent. This is a contradiction, including when $k = 2$. $\square$

Corollary 6.2. *Under the hypotheses of Theorem 1.1 or Corollary 1.2, a paired clique model can be found in polynomial time when the regions are supplied.*

Proof. For the symmetric statement, make the terminal assignment from the proof of Corollary 1.2; for $k = 1$, return a path. It remains to give a construction under the one-sided hypotheses. Vertex-capacitated maximum flow finds a linkage of order k and tests whether one survives a specified contraction.

First absorb or delete components outside R and the regions, as in Section 2. If a region is a singleton $\{s\}$, take a linkage of order k and let r be the other end of its path ending at s . Recurse after deleting r, s and this region, then append the branch set $\{r, s\}$. This recursion has $k - 1 \geq 2$ and satisfies the smaller theorem's hypotheses.

Otherwise test every permitted region edge. If contracting one preserves a linkage of order k , recurse on the quotient and lift the returned branch sets through that edge. If all these tests fail, return any linkage of order k . Its paths are pairwise adjacent by Sections 3 through 5: after normalisation and removal of the singleton case, those sections use minimality only through (2). The algorithm therefore does not construct the cut lattice or enumerate its maximal chains.

Every recursive call decreases the number of vertices. There are therefore at most $|V(G)|$ calls, each using polynomially many flow tests and elementary graph operations. The disjoint connected preimages used for every contraction lift are recorded during the recursion. Component deletions and the singleton step require no further allocation, so all lifts also take polynomial time. $\square$

REFERENCES

- [1] K. Menger, *Zur allgemeinen Kurventheorie*, Fund. Math. **10** (1927), 96–115.
- [2] E. Protopapas, D. M. Thilikos and S. Wiederrecht, *Colorful Minors*, arXiv:2507.10467v3, 2026.